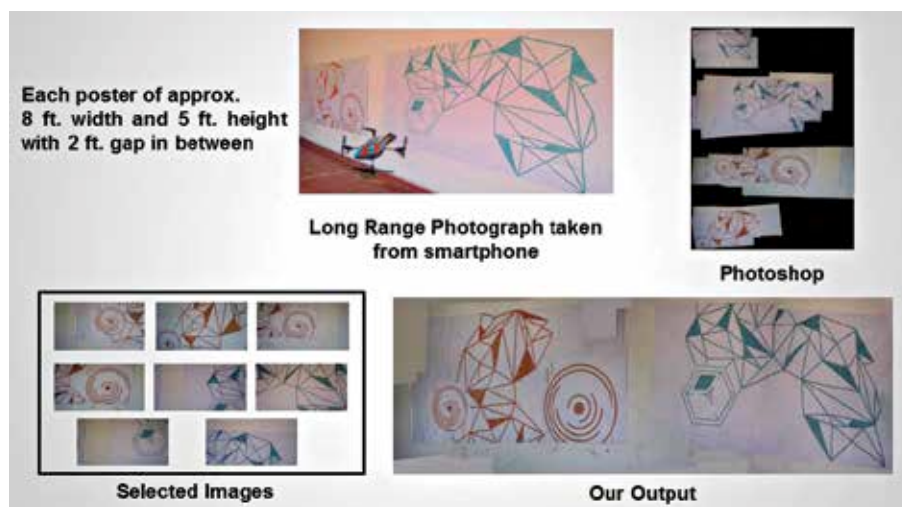




A demonstration of the panorama creating technique

duced due to jerky motion of inexpensive quadcopters. We have designed blur resilient fiducials. So, not only can each quadcopter identify other quadcopters, we can even instruct a quadcopters according to the code embedded in the fiducial on it.”

While imaging for personal consumption or inspection has its uses, they have also considered an important public health issue. One of the major reasons for the dengue health hazard is the number of locations where one can find stagnant water – usually found in uncovered gutters or terraces of high rise buildings, or other areas that are hard to reach and access. The team has proposed the use of a quadcopter to inspect such areas and identify stagnant water patches using a novel optical flow technique.

A LEAN MOOC PLATFORM

Software and programming developments over the last two decades have changed the Earth as we know it. One such development is the emergence of MOOC (Massive Open Online Course) platforms. The New York Times labelled 2012 as the year of the MOOC. One of the problems with MOOCs is their inability to cope with low-bandwidth and low-computing power situations. In an attempt to solve for this problem, a team at IIT Kanpur are working on mooKIT – a MOOC management system being worked on by Prof. T V Prabhakar and his team.

Existing MOOC software (such as openEdx) are complex to host, manage and adapt to local needs. mooKIT aims to change that. It is targeted specifically at developing nations, and has already been

used in 12 courses in India and abroad. The most recent of those is a MOOC on History and Philosophy of Science by the National Open University of Nigeria. mooKIT has also been adopted to run courses in the blended mode (Flipped classes) at IIT Kanpur.

The connectivity challenges of users in a developing country include frail, unstable connections that aren't able to stream video, and are expensive to boot. mooKIT has an indicator showing the current bandwidth of the internet connection, quite like the bars we have on a cell phone. This gives a visual indication to the student if the connection is bad, and she can use other content delivery options that mooKIT provides – for example, stream only audio and play it in sync with the slides, which is often very close to the video experience. If the



Professor PK from the Precog division!

bandwidth is too low for that even, you can receive a call on the phone and listen to the audio from there!

A unique feature is the integration with social networking platforms such as Facebook and Twitter. All discussions in the course automatically get posted on a FB group, and also go out as tweets.

A very powerful analytics interface is available to give insights on course activities. Instructors get sophisticated information related to the level of involvement of each student, and also get predictive assessments related to these. Sentiment analysis can be performed on forum posts to analyse the satisfaction of students.

mooKIT is built entirely using open source technologies, and the core engine runs on the MEAN stack, making it extremely scalable.

FIND MALICIOUS FB PAGES

Facebook is home to your friends and family, and also to hostile entities who post malicious content. Because Facebook is such a vital part of our lives, we need to identify and stop them.

Precog is a group of researchers from IIT Delhi who work on different aspects of social networking (yes, the name is intentionally from *Minority Report*!). Praatek Dewan, Ph.D Student, Shrey Bagroy, an Undergraduate student, and Prof. Ponnuram Kumaraguru (PK) from the group have worked on identifying such malicious pages that published untrustworthy information, misleading content, adult and child unsafe content, scams, etc., and have submitted their findings to at IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM) 2016, where it has been accepted.

“We scanned close to 400 thousand Facebook pages and leveraged Web of Trust's (WOT) crowdsourcing mechanism to identify pages which engaged in spreading any such malicious content. We identified over 600 Facebook pages which posted malicious content at least once in their one hundred recent posts. Upon comparing the behaviour of these 600 odd pages with an equal random sample of benign pages, we noticed that malicious pages were not only more active, but also very well interconnected within themselves as compared to benign pages.”, explains Prof. Ponnuram. He adds,